LungMarker: results summary

Research use only. Not for clinical use. No clinical claims. AI-assisted finding, measuring and matching of lung nodules on CT, built for the Harbor Thoracic Oncology Lab (simulated). Tested on public data (RIDER Lung CT); not yet on the lab's own scans. Status 19 Oct 2026. Full version: `RESULTS.md` in the repository.

1 · TIME

Estimate: the tool's own mechanics take about 1–2 minutes per scan; the reader's judgement time with the tool is not measured yet, so the time saved is still a range.

- Today, by Daniel's own account, a follow-up scan takes **40–55 min** by hand (two-thirds of incoming scans are follow-ups).
- Developer pilot, 7 scans: opening, confirming, editing, matching and export took **1–2 active min**. That is overhead, not reading time.
- With the tool, a follow-up takes that overhead plus Daniel's checking of each AI draft: somewhere between a few minutes and close to today's time. Whether the lab's target (about half) is met can't be shown yet.

2 · QUALITY

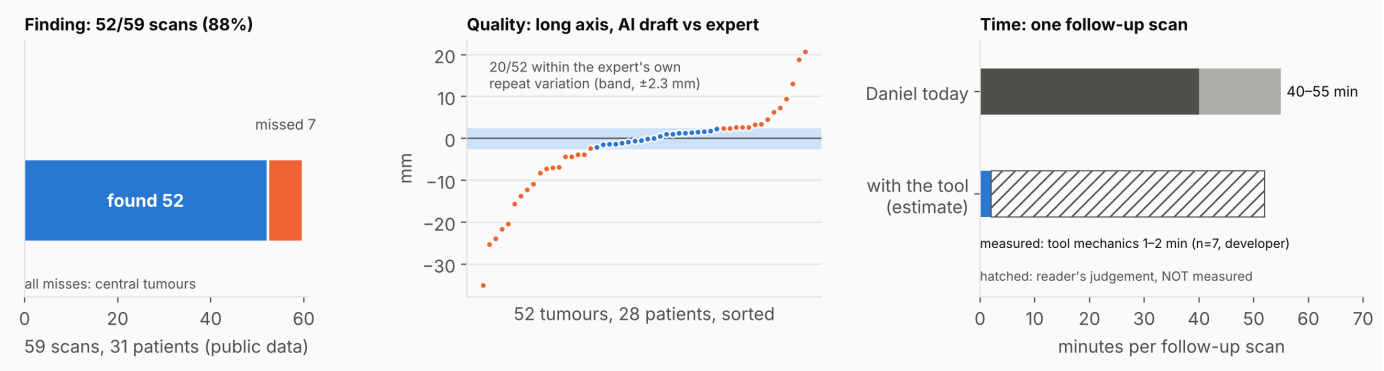
On 52 tumours from 28 patients, the AI's first draft is close to the expert on average but not as precise per tumour, so every value is checked and confirmed before it is recorded.

- Expert-marked tumour found in **52 of 59 scans (88%)**; matched to the earlier scan in **24 of 24** pairs.
- Long axis: median error **3.0 mm**, no systematic bias; **20 of 52** within the expert's own scan-to-scan variation (2.3 mm). Volume: median error 14%; 27 of 52 within 17%.

3 · FAILURES

The AI misses some central tumours and draws some outlines far off; the app makes the reviewer catch these, and some cases are untested.

- All 7 misses (4 of 31 patients) are central tumours touching the hilum or mediastinum, so the reviewer must check those areas before approving.
- 19 of 52 long axes are more than 5 mm off the expert. Not tested yet: faint ground-glass nodules, false alarms, the lab's own scans.



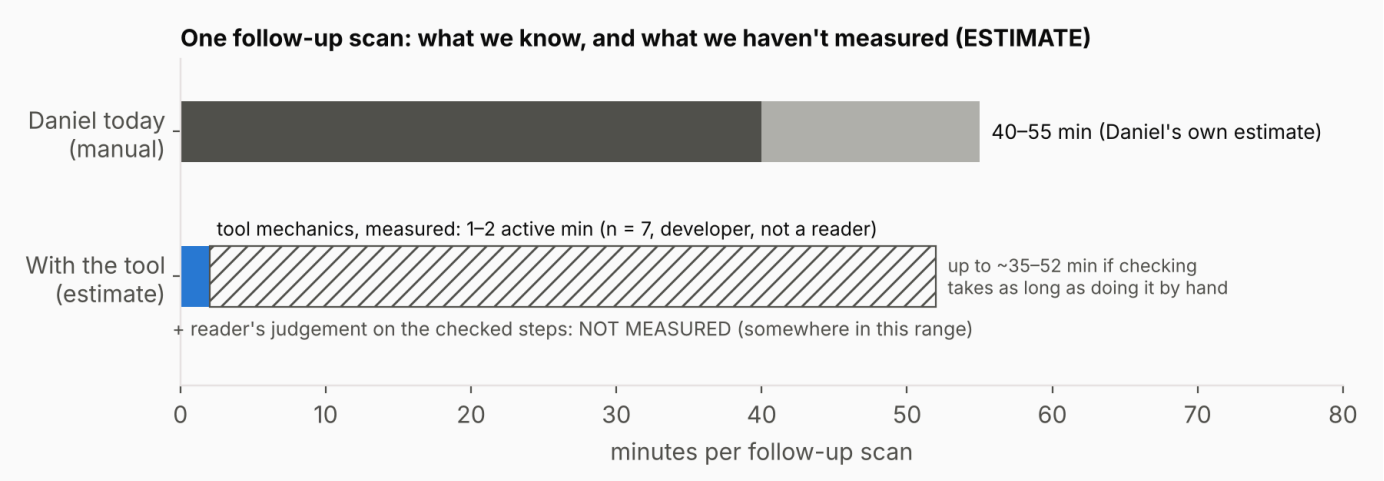
Left: tumours found out of 59 scans. Middle: each dot is one tumour's long-axis error against the expert; the blue band is the expert's own variation between two scans of the same patient. Right: one follow-up scan today, and with the tool, where only the tool's mechanics are measured (blue) and the reader's judgement time is not (hatched). All on public data; research use only.

How time was measured

Why an estimate. The real test is Daniel timing his own work with and without the tool on the lab's scans. That isn't possible yet: the scans never leave the hospital and the tool runs on public data. So, as agreed with Base44's FDE lead (Sam, 18 Oct), we combine **Daniel's step-by-step account of his manual review** with **the developer's timed runs in the tool**, and label everything as an estimate.

Step (follow-up scan)	Daniel today	What the tool does	Effect	Left with the tool
Loading	1–2 min	Opens with the AI's marks ready; prior scan already linked	Shortened	Overhead (measured)
Re-find and match each prior nodule	~10 min per nodule (aligns by bronchi and fissures)	Suggested match; prior and current side by side	Shortened	His check: not measured
Find new nodules	10–20 min (two passes, 250–400 slices)	AI marks candidates; one check pass + required central check	Search shortened; check kept	Not measured
Measure	2–4 min per nodule	Long and short axis drafted	Shortened	Not measured
Full outline	15–25 min per tumour, when needed	Drafted for every nodule; volume for every confirmed one	Shortened; more nodules	Not measured
Recording	3–5 min	One Excel export a day, in his sheet's format	Removed	Overhead
New: decide on each AI mark; central check	—	Accept or reject every suggestion; tick "hilum and mediastinum checked"	Added	Not measured
Total	**40–55 min**			**Overhead 1–2 min + judgement (not measured)**

Daniel's steps are in his own words (Slack, 18 Oct). Some are per nodule, so the total is his own figure, not our sum; his log (one follow-up, 55 min) agrees within 14%. A baseline scan takes about 39 min (log, 4 scans).



Developer pilot: 7 scans (4 patients, baseline and follow-up), 1 more excluded (left open while doing something else), 2–5 nodules each, active minutes from the app's timer, rounded to whole minutes. Every run is listed in the appendix.

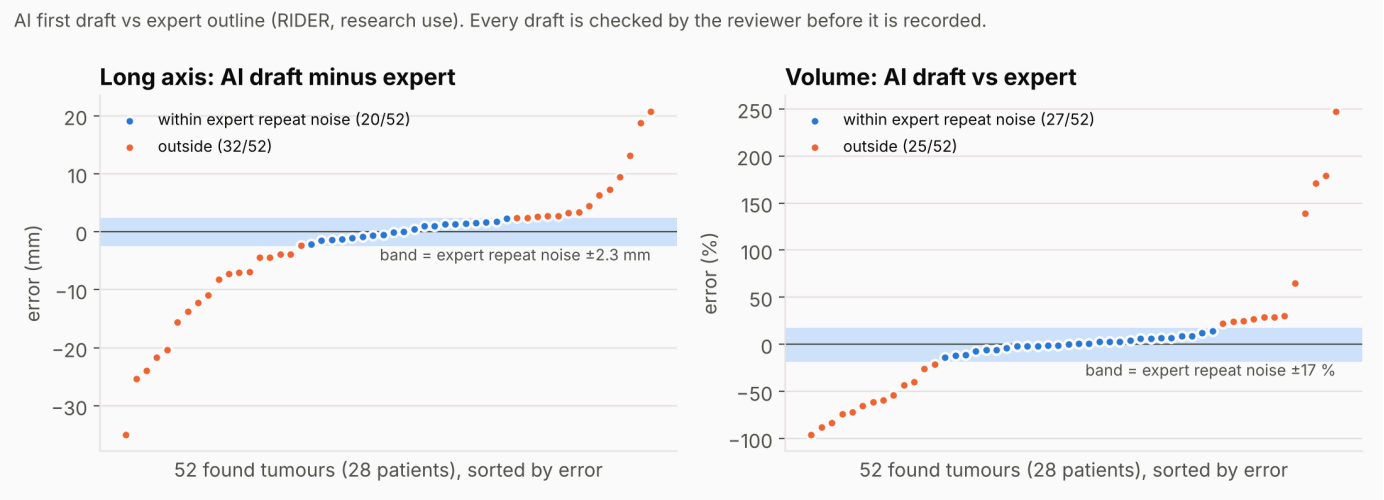
How to read the range. Low end: the mechanics alone (~2 min), not a realistic reading time. High end: today's time minus recording (~35–52 min), as if checking each AI draft took as long as doing the step by hand; the new steps could push it higher. We report no single "with the tool" figure and no percentage saved: these data can't support one. Main caveats: the developer is not a trained reader, and the public scans have larger, clearer tumours than the lab's.

Quality against the expert

Test set. RIDER Lung CT (public): 31 patients scanned twice, 15 minutes apart; 59 scans with an expert outline of one tumour each. The pipeline ran with no hint. Every measurement is taken the way Daniel takes it (long axis on the slice where the tumour is largest, short axis at right angles, volume from the outline), on the AI's outline and on the expert's.

What was compared	Sample	Result
Finding: expert tumour found, no hint	59 scans, 31 patients	**52 / 59 (88%)**
Long axis: AI draft vs expert	52 tumours, 28 patients	median error **3.0 mm** (bias −0.3 mm); **20 / 52** within 2.3 mm
Volume: AI draft vs expert	52 tumours, 28 patients	median error **14%** (bias +0.5%); **27 / 52** within 17%
Matching to the earlier scan	24 patient pairs	**24 / 24** correct
Lobe	52 tumours	52 / 52 agree with the expert outline
Small nodules (< 8 mm), same nodule on both scans (no expert outline)	84 pairs, 22 patients	diameter differs by a median 0.5 mm; volume by ~30%

Yardstick: the expert's own variation between the two scans of the same patient (median 2.3 mm long axis, 17% volume; 28 patients).



- **What this measures:** the AI's first draft. What goes into the lab's sheet is the reviewer's confirmed value, which was not measured here; the edit trail records every change, so it can be measured in real use.
- **The test tumours are large** (13–93 mm, median 43 mm); the lab measures from 4 mm, so small-nodule accuracy is not established.
- **Somewhat optimistic:** the outline settings were compared on these same 52 tumours.

Failure cases

Failure	How often	What the app does about it
Central tumours missed (touching the hilum or mediastinum), even large ones, 14–52 mm	7 / 59 scans, 4 / 31 patients; all of the AI's misses	Approve stays blocked until the reviewer ticks "I checked the hilum and mediastinum" (one click to the mediastinal window; quick Add nodule). An automatic "check here" flag was tried and dropped: it caught 4 of the 7 but flagged 49 of the other 52 scans.
Outline far off: long axis more than 5 mm from the expert	19 / 52 tumours (14 patients): 13 too small, 6 too large	Nothing is recorded until the reviewer confirms the outline; Erase, Draw and point editing fix it. The 4 small faint tumours drawn much too small now show "AI unsure about the edges"; the 9 large ones are not flagged.
Small-nodule volume is approximate	~30% between repeat scans for nodules < 8 mm (84 pairs)	Diameters are the reliable number for small nodules; volume is recorded only after the reviewer confirms the outline.
Wrong follow-up match (would also misalign the side-by-side view)	Not seen (24 / 24 correct), but possible	The match is a suggestion the reviewer picks; a warning flags a different lobe, a size change over 50%, or disagreement with the alignment.

Limits

- **Not tested:** faint ground-glass nodules (none in the test data; Daniel's hardest cases), false alarms (the data marks one tumour per scan; only a proxy: 82% of AI marks reappear on the repeat scan), and the lab's own scans.
- **Public practice data:** mostly large, solid tumours; the lab's cohort may differ.
- **First draft only:** we measured the AI's draft, not the reviewer's confirmed values.
- **Same tumours for tuning and testing:** outline settings were compared on the 52 test tumours, so quality numbers are somewhat optimistic.
- **Follow-up = same-day repeat scan:** tests matching, not growth.
- **Timing is an estimate:** small samples (7 runs, a handful of log rows), a developer rather than a reader, and the reader's judgement time not measured.

Next steps

- **Measure the reader's judgement time:** Daniel times his own reviews with and without the tool on a small set of the lab's scans, in a hospital pilot with the engine on one hospital server. This is the only way to turn the timing range into a number, and it also measures the quality of his confirmed values.
- **Ground-glass test** on the public LNDb set (27 ground-glass + 46 part-solid nodules ≥ 4 mm).
- **Central tumours:** try a second model trained on lung-cancer outlines; keep the manual central check until a test shows it works.
- **Follow-up view:** export the lung alignment from the engine, so matching checks and the side-by-side view use it.
- **Cost for a research budget** (last year's tool failed on price): write down the full yearly cost before any rollout, namely the Base44 plan, one hospital server (an existing machine if it has a GPU or at least 8 GB of memory) and the IT hours, so the lab can compare it with last year's tool. No licence or per-scan fees for the AI models.

Why this differs from last year's tool

Why it failed	What LungMarker does	Honest caveat
Accept / reject only, no editing of outlines	The reviewer edits the outline directly (points, Erase, Draw); volume only after Confirm outline; every change kept in an edit trail (AI draft, final value, who, when).	Tested by us, not yet by Daniel on his own scans.
No export to their sheet	One-click Excel export in the lab's exact 18-column format, plus the edit trail. The sheet stays the single source of truth.	Checked on real exports and 30 automated checks, on public scans.
Needed hospital IT to install	Runs in the browser; nothing installed on anyone's computer.	Some IT remains: the AI engine must run on one hospital server (batch, from the image archive), plus a data-governance review of what the cloud app stores (study IDs, measurements, edit trail; no images).
Too expensive	Open-source AI models (no licence or per-scan fees) on a low-cost app platform.	Yearly cost not yet written down (next step).

Appendix A · Every timed run (developer pilot, 18 Oct, final app version)

Patient (public RIDER)	Timepoint	Active min	Open min	Nodules reviewed	Note
RIDER-1225316081	Baseline	1	1	2	
RIDER-1225316081	Follow-up 1	*12*	*20*	3	**Excluded:** scan left open while writing messages
RIDER-1332496276	Baseline	1	1	2	
RIDER-1332496276	Follow-up 1	1	1	2	
RIDER-1500037140	Baseline	2	2	4	
RIDER-1500037140	Follow-up 1	2	2	4	
RIDER-1532432635	Baseline	2	2	5	incl. 2 rejected
RIDER-1532432635	Follow-up 1	2	2	5	incl. 1 rejected + 1 not found
Included: 7 scans		**1–2 (median 2)**		**2–5**	**developer, not a reader**

Source: the app's own timer ("Minutes on scan" and the approval rows of the edit trail in the Excel export), active minutes: paused after 2 min without input or when the tab is hidden; whole minutes, so "1" means roughly 0.5–1.5 min.

Appendix B · Daniel's manual baseline (his words, Slack, 18 Oct)

Per follow-up scan: loading 1–2 min; re-finding and matching ~10 min per nodule (aligns by bronchi and fissures; slice numbers are unreliable because of breathing); finding new nodules 10–20 min (two passes over 250–400 slices); measuring 2–4 min per nodule; full outline 15–25 min per tumour when needed; recording 3–5 min; total 40–55 min. His log ("Minutes on scan", last 5 rows): baseline 38, 49, 30, 37; follow-up 55. About two-thirds of incoming scans are follow-ups (his estimate).

Appendix C · Technical method

- **Finding:** TotalSegmentator `lung_nodules` (nnU-Net, ~1,350 training CTs partly from LIDC-IDRI, not RIDER; Apache-2.0) at full settings, on a free Colab T4 GPU (median 49 s per scan). Lobes: TotalSegmentator `total`.
- **Outline:** the model's core plus a rim scaled to the core's size (shipped setting). Compared first with the model's own outline and MedSAM 2; four rim variants compared on the 52 tumours (`halo_variants_found52.csv`).
- **Measurements:** long axis = longest diameter on the slice where it is largest; short axis = longest perpendicular on that slice; volume from the 3D outline; lobe = majority lobe.
- **Matching:** affine lung registration (SimpleITK) + Hungarian pairing, gate max(10 mm, 0.75 × long axis).
- **Expert yardstick:** median absolute difference between the expert's manual outlines on the two scans of each patient (28 patients).
- **"AI unsure about the edges":** mean model probability over the outline < 0.6, or the rim would grow the long axis by ≥ 4 mm.

Appendix D · Raw data and reproduction

All in the engine repository, `results/`: `finding_exam_all59.csv` (one row per scan), `finding_exam_matching.csv` (per patient pair), `halo_variants_found52.csv` (outline settings; shipped = `scaled_*`), `small_nodule_repeatability.csv`, `expert_baseline.csv`, `central_check_blob_all59.csv`, `faint_flag_candidates.csv`, `daniel_steps.csv`, `time_log.csv`. `python results/make_figures.py` rebuilds every figure and prints every number in this report; `python results/build_pdf.py` rebuilds this PDF. Data: RIDER Lung CT (Zhao et al., 2015) and RIDER-LungCT-Seg (Wee et al., 2020), The Cancer Imaging Archive.